Dell CloudLink

SNMP Reference Guide

October 2025
Rev. 1.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Revision History

Table 1. Revisions

Date	Document Revision	Description of Changes
October 2025	1.0	Initial release for CloudLink 8.2.x.

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About Dell CloudLink SNMP

The Dell SNMP reference guide provides information about CloudLink with the Simple Network Management Protocol (SNMP) support features, traps, and the Management Information Base (MIB).

The guide also describes how to manage SNMP in the CloudLink Center.

The audience for this guide are CloudLink system administrators who:

- Use SNMP manager.
- Troubleshoot alarms in CloudLink Center.

Dell CloudLink supports the following SNMP versions:

- SNMPv2
- SNMPv3

CloudLink MIB

The CloudLink DELL-CL-SURVEILLANCE-MIB provides information about the traps that are sent by CloudLink Center. SNMP OID 1.3.6.1.4.1.3183.1.1. identifies the CloudLink MIB.

Each alarm, contains the following fields:

- System Up Time
- SNMP Trap OID
- Alarm ID
- Alarm Type
- Alarm Name
- Alarm Time Stamp
- Alarm Severity
- Alarm Target Type
- Alarm Target Name
- Alarm Details

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Configure SNMP for CloudLink Center

This section provides info about adding, configuring, and test SNMP traps in CloudLink Center.

CloudLink Center alarms as SNMP traps

Alarms are activated in response to an event or a set of conditions. For example, CloudLink Center triggers an alarm if a backup file is generated but not downloaded. When you download the backup file, the alarm is automatically cleared.

Alarms have High and Low severity levels.

CloudLink Center sends SNMP traps to a single target when CloudLink alarms are triggered, updated, or cleared.

Because cluster members do not synchronize SNMP configuration, each server in a CloudLink cluster must be configured separately. The following traps are sent from CloudLink Center:

- `alarmUpTrap`—When a new alarm is triggered on the CloudLink Center server.
- `alarmDownTrap`—When the cause of an alarm is corrected or resolved.

Information about the CloudLink alarm is stored in the alarmUpTrap, and the alarm identifier is stored in `alarmDownTrap`.

Add an SNMP configuration to CloudLink Center

Add an SNMP configuration to the CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **SNMP** > **Add**.
3. From **Add New SNMP Configuration**, in the **Target Version** list, select **SNMPv2** or **SNMPv3**.
   - If you select SNMPv2, then enter:
     - **Host**—The IP address or FQDN to where the SNMP traps must be sent.
     - **Port**—The receiving host port number.
     - **Description**—Description for the selected target.
     - **Community**—The SNMP trap community.
   - If you select SNMPv3, then enter:
     - **Host**—The IP address or FQDN to where the SNMP traps must be sent.
     - **Port**—The receiving host port number.
     - **Description**—Description for the selected target.
     - **Engine ID (HEX)**—The remote SNMP protocol engine identifier. This value must be between five and 32 octets in length.
     - **User Name**—The username is required to communicate with the SNMP agent.
     - (Optional) **Authentication Type**—Specifies the type of authentication used.
     - **Authentication Password**—Password used to authenticate a connection with an SNMP agent. Must be of at least eight characters in length.
     - (Optional) **Privacy Type**—Specifies the type of encryption used.
     - **Privacy Password**—Password used to encrypt a connection with an SNMP agent. Must be at least eight characters in length.
4. Click **Add**. The SNMP configuration is added to CloudLink Center.

Modify the SNMP configuration in CloudLink Center

Modify the SNMP configuration in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP**.
3. Select the SNMP and click **Modify**.
4. In the **Modify SNMP Configuration** dialog box, modify the required values, and then click **Modify**.

Send a test SNMP trap in the CloudLink Center

Test your SNMP configuration to ensure that the target device receives SNMP traps from the CloudLink Center.

About this task

Following traps are sent from CloudLink Center:

- alarmUpTrap when a new alarm is raised on the CloudLink Center server.
- alarmDownTrap when the cause of the alarm is corrected or removed.

The information about the CloudLink alarm is contained in the alarmUpTrap, and the alarm identifier is contained in the alarmDownTrap.

You can use the clAlarmId for unique value for each alarm instance to track the raising and lowering of a specific alarm on the CloudLink Center server.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP** > **Send Test Trap**.
3. In the **Confirm Sending Test SNMP Trap** dialog box, when prompted to confirm the request, click **Send**.

Download MIB files

Import or upload the MIB file to the trap receiver or network management system application. CloudLink alarms use the Object Identifier (OID) 1.3.6.1.4.1.1139 for traps.

About this task

To download MIB files:

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP** > **Download MIB**.

Delete the SNMP configuration in the CloudLink Center

Delete the SNMP configuration in the CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP**.

3. Select the SNMP host, and click **Delete**.
4. In the **Confirm SNMP Configuration Deletion** dialog box, when prompted to confirm the request, click **Delete**.

View CloudLink Center alarms

An alarm represents a state or condition. When one or more alarm conditions exist, CloudLink Center displays a badge on the **Alarms** icon on the home page. The badge indicates the number of alarms.

Steps

1. Log in to CloudLink Center.
2. On the **Home** page, click  .

Next steps

CloudLink Center removes alarms from the **Alarms** page when the condition or state that triggered the alarm is resolved. For example, CloudLink Center triggers an alarm if a backup file is generated but not downloaded. When you download the backup file, the alarm is automatically deleted.

Change the CloudLink Center alarm state

CloudLink Center provides two alarm states: Watched and Ignored.

About this task

To change the alarm notification states and email notification status in CloudLink Center:

Steps

1. Log in to CloudLink Center.
2. Select **System** > **Alarms Configuration**.
3. Select the alarm you want to change.
4. Select **Actions** > **Change**.
5. In the **Change**, dialog box, select the following:
   - From the **State** list, select one of the following values:
     - **Watched**—Future notifications for the selected alarm are reported.
     - **Ignored**—Future notifications for the selected alarm are ignored.
   - From the **Email Notification** list, select one of the following:
     - **Send**
     - **Do not Send**
6. Click **Change**.

CloudLink Center alarms

Provides a list of CloudLink Center alarms, causes, and relevant actions.

Table 2. CloudLink Center alarms and relevant actions

Sl. No.	Alarm	Description	Cause	Severity	Action
1	The keystore is not accessible	CloudLink Center keystores are not reachable. CloudLink Center supports the	The CloudLink Center keystore locations are not reachable.	High	This alarm is cleared when the keystore locations are accessible. No action is required.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
		following keystore locations: ● Local database ● Microsoft Active Directory ● Amazon S3 ● S3-compatible bucket			
2	The cloud provider is not accessible	The Cloud service provider is not reachable. CloudLink Center supports the following cloud service providers: VMware vCenter, Microsoft Azure, and Amazon Web Services.	Incorrect information is entered while adding a Cloud service provider.	Low	Ensure that you enter the correct information while adding a Cloud service provider.
3	Instance-based license violation	The instance-based license compliance requirement is violated.	The number of machines: virtual or bare-metal that is protected by the CloudLink Center is greater than the specified number of machines in the instance-based license.	High	To clear this alarm: ● Remove the extra machines than the specified limit permitted in the instance-based license. ● Upload an instance-based license for the extra machines.
4	Capacity-based license violation	The capacity-based license compliance requirement is violated.	The storage capacity of the license reaches its threshold value, and a storage node device is added which collectively exceeds the storage limit of the capacity-based license.	High	To clear this alarm: ● Remove a few SDS devices to reduce the storage capacity. ● Upload capacity-based license for the exceeded storage capacity.
5	KMIP license violation	The KMIP-based license compliance requirement is violated.	The number of KMIP clients is greater than the KMIP clients limit allowed in a KMIP-based license.	High	To clear this alarm: ● Delete the extra KMIP clients. ● Upload a KMIP-based license.
6	SED-Instance-based license violation	The Self-Encrypting Disk (SED) instance-based license compliance requirement is violated.	The total number of SED drives in physical machines exceeds the specified limit in the SED instance-based license.	High	To clear this alarm: ● Remove few machines that contain SED drives. ● Upload SED-instance-based license.
7	Domain controller is not accessible	The Microsoft Windows domain controller is not accessible.	This alarm is triggered when: ● The LDAP protocol and the Global Catalog port are unreachable. ● Incorrect information	High	To clear the alarm: ● Enter correct information while configuring Windows Domain Controller. ● Ensure that the LDAP and Global Catalog port are accessible.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
			is entered while configuring Windows Domain.		
8	Domain controller configuration in progress	An AD configuration is in progress.	This alarm is triggered when: • CloudLink Center is restarted. • AD configuration is changed.	High	This alarm is cleared when the AD configuration is completed. No action required.
9	CloudLink Vault is locked.	CloudLink Vault is locked. You can choose to set the CloudLink Vault modes as follows: **Auto Unlock** or **Manual Unlock** while configuring CloudLink Center.	This alarm is triggered when: • The Manual Unlock feature is selected while configuring the CloudLink Center server. • The Manual Unlock requires a passcode.	High	Enter the passcodes to unlock the CloudLink Vault.
10	CloudLink Vault unlock codes are not configured	CloudLink Vault unlock codes are not configured.	CloudLink Vault passcodes are not set while configuring the CloudLink Center.	High	Ensure that you enter the CloudLink Vault passcodes while configuring the CloudLink Center.
11	Failed to generate new backup file	CloudLink Center is unable to generate a new backup file.	Database size is larger than the /tmp directory of CloudLink Center.	High	Contact the Dell Technologies Support team to reduce the database size by deleting few records or files.
12	Failed to upload backup file to store	CloudLink Center fails to upload the backup file to a backup store. CloudLink Center supports the FTP, SFTP, FTPS, Amazon S3 Storage, and S3 Compatible Storage backup stores.	CloudLink Center cannot write the automatically created backup file to the target storage.	Low	To clear this alarm, verify the: • Target data store accessibility. • Target data store user credentials and permissions. • Target data store ciphers.
13	Backup file was not downloaded or uploaded to the store	Backup file was not uploaded or downloaded to a backup store on a scheduled basis.	This alarm is triggered when: • You did not use a backup store to upload the backup file in the last seven days. • You did not manually download the backup file in the last seven days.	High and Low	Ensure that you upload or download the backup file.
14	Backup encryption key was not generated or downloaded	Backup encryption key pair was not generated and downloaded when CloudLink Center is	Backup key pair was not generated and downloaded when the CloudLink Center	Low	Generate and download the backup key pair.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
		configured on Microsoft Azure. A backup is stored in a file that is encrypted using an AES-256 key that is protected by an RSA-2048 key pair.	is configured on Microsoft Azure.		
15	Pending Machine	When a registered machine with an encrypted boot volume or a device attempts to start up, CloudLink Center cancels the startup process and puts the machine in the pending state. This state implies that key release policies are not met, and CloudLink Center does not release encryption keys to the boot volume or the device of a machine. The pending machine policy determines if a machine should be placed in the pending state when it connects to CloudLink Center for the first time. Or, when it connects from a network or location that is not previously associated with the machine.	The pending machine policy is set to **Require Manual Approval**.	Low	Accept the pending machine.
16	Machine is waiting for key	Machine is waiting for the keys to be released from the keystores. When the boot or data volumes of a machine is encrypted, keys are generated and stored in CloudLink Center keystores. During the startup process of a machine, the keys that are stored in the keystores are released, and then the machine connects.	The keys that are stored in the keystores are not released during the startup process of a machine.	Low	To clear this alarm, verify if: • External keystore is accessible. • The key is in the external or local keystore. • CloudLink cluster is synchronized. Restore keystore from the backup file.
17	Volume encryption policy violated	Volume of a machine does not comply with the volume encryption policy of a machine group.	The volume encryption policy for a machine group applies to all the volumes of machines associated with a particular machine	Low	CloudLink Center displays an exempt link next to the volume name on the **Machines** panel. To clear this alarm, specify that you want to allow

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
			group. You may want to exempt a volume from the policy. For example, you might have assigned the Boot and All Data volume encryption policy to a machine group. However, you want one volume on an individual machine in that group to be decrypted. When you manually decrypt this volume, it no longer complies with the volume encryption policy for that machine group.		the noncompliance by exempting the volume.
18	Uncontrolled BitLocker protector detected	The volumes of a drive are not controlled by CloudLink. Few drives are auto-protected and do not require encryption from CloudLink.	Volumes of a drive are either in Protected or Uncontrolled state.	Low	To clear this alarm: • Delete the machine with the drives that are auto-protected. • Change the alarm state to **Ignored**.
19	Machine reports newer version	A VM installed with the latest version of CloudLink Agent tries to connect to an older version of the CloudLink Center. The version of CloudLink Agent must be the same as or earlier than the CloudLink Center version.	CloudLink Agent tries to connect to an older version of CloudLink Center.	Low	To clear this alarm: • Upgrade the CloudLink Center to match the CloudLink Agent version. • If all drives are unencrypted, uninstall the CloudLink Agent, and then install the CloudLink Agent that has the same version of CloudLink Center.
20	Machine upgrade is required	When upgrading CloudLink Center, the connected CloudLink Agents are automatically upgraded if the Machine Agent Upgrade policy is set to **Auto**.	This alarm is triggered when the: • The Machine Agent Upgrade policy is set to **Manual**. • The CloudLink Agent is not upgraded even though CloudLink is upgraded to a newer version.	High	Manually upgrade the CloudLink Agent after you upgrade CloudLink.
21	Cluster member state error	The CloudLink Center cluster displays erroneous states for the Incoming or Outgoing status of the cluster nodes.	The incoming or outgoing statuses of the CloudLink Center cluster node are in the following erroneous state:	High	This alarm is cleared when the incoming or outgoing status of the CloudLink Center cluster node is not in an erroneous state.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
			• ERROR • OFF • TIME DIFFERENCE		No action required.
22	KMIP Client authentication is not successful	Username and password are provided while adding a KMIP client in CloudLink Center. KMIP clients are accessed using this password.	Incorrect password is entered while accessing the KMIP client.	High	Ensure that you enter the correct password to access the KMIP client. If you have forgotten the password, change the KMIP client password.
23	One or more keys are not found	CloudLink Center checks if it can access volume or device keys for the connected and registered machines on a scheduled basis.	This alarm is triggered when the: • CloudLink Center cannot access the volume or device keys that are stored in the keystore because the keystore containing the required key is not accessible. • Keystore that contains required volume or device keys is deleted. • Key from a keystore is deleted during the key shredding process or by using external tools.	High	Check the network connectivity for the keystore or restore a keystore from the backup file.
24	Failed to send notification email	CloudLink Center is unable to send the email notification whenever an alarm is triggered or updated.	Email notification is not sent to the configured SNMP targets.	Low	This alarm is cleared when the specific email notification is sent to all the configured SNMP targets. No action required.
25	Failed to send SNMP trap	SNMP trap is not sent to the configured SNMP targets.	SNMP trap is not sent to the configured SNMP targets.	Low	This alarm is cleared when the specific SNMP trap is sent to all the configured SNMP targets. No action required.
26	Unknown machine location	Machine is not located in an approved location. Each machine that is registered with CloudLink Center belongs to a machine group. You can create an approved location and select it as an approved	The location of the machine is changed and is not operating in an approved location.	High	This alarm is cleared when the client VM is migrated to an approved location. No action required.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
		location for multiple machine groups.			
27	Untrusted machine location	The VMs can be moved from one location to another within the approved Cloud service providers. A Cloud Provider instance is a location in a Cloud service provider, such as a data center or a cluster. The instance can be defined as a Cloud service provider or a subset of a Cloud service provider.	The machine is moved from a Cloud Provider instance to an untrusted location.	High	To clear this alarm: ● Migrate the machine to an approved location. ● Add the machine location to approved locations.
28	KMIP Client Certificate is invalid	The KMIP client certificate is invalid. Adding a KMIP client also generates a certificate for the KMIP client with an expiration date.	This alarm is triggered when the: ● Validity of the KMIP client certificate in CloudLink Center has expired. ● Certificate Authority (CA) manager failed to verify the KMIP client certificate.	High	To clear this alarm: ● Delete the KMIP client whose certificate is invalid. ● Install the new certificate on the KMIP client.
29	KMIP CA Certificate is expired	The KMIP CA certificate is used to sign the server and client certificates.	This alarm is triggered when the validity of the KMIP CA certificate in CloudLink Center has expired.	High	To clear this alarm: ● Regenerate or upload a third-party KMIP CA certificate. ● After generating or uploading the CA certificate, regenerate the KMIP client certificate.
30	System does not meet the minimum requirements	The minimum system requirements for deploying CloudLink Center are four vCPUs and 6 GB vRAM.	The available number of vCPUs and vRAM space is not as per recommendations.	High	Ensure that your system meets the minimum requirements for deploying CloudLink Center.
31	System database exceeds warning threshold	The size of the CloudLink database has exceeded the specified limit.	The size of the CloudLink database has exceeded the specified size limit of 2 GB.	High	Contact the Dell Technologies Support team to reduce the database size by deleting few records or files.
32	Available disk space is below safe threshold	The CloudLink Center Linux VM has " /" partitions in the root directory. CloudLink Center defines a maximum space limit of 75% for each "/" partition.	The "/" partition free space is less than the 25% threshold limit.	High	To clear this alarm: ● Replace the CloudLink cluster node with a new node. ● Contact the Dell Technologies Support team to free the partition space.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
33	Volume pending	The Kubernetes volume is in pending state.	This alarm is triggered when the: • Volume is initialized, and the Kubernetes nodes are restarted. • Volume is initialized in one node, and the other node tries to access the volume simultaneously.	Low	Accept the pending Kubernetes volume.
34	SDS Device header is unavailable	When an SSD is encrypted, a Storage Data Server (SDS) device header is generated indicating that PowerFlex is using a raw device.	The SDS device header is not added in the PowerFlex when an SSD is encrypted.	High	To clear this alarm: • Remove the SSD from PowerFlex. • Restart the CloudLink agent.
35	CloudLink Agent is no longer supported on CoreOS	CoreOS does not support installation of CloudLink Agent from CloudLink 7.0 and later versions.	When a machine running on CoreOS using CloudLink 6.9 or earlier versions is upgraded to CloudLink 7.0 versions or later.	High	Disconnect or remove the CloudLink Agent installed on CloudLink 6.9 or earlier versions running on CoreOS.
36	Duplicate IP for machine	Two machines with the same IP address exist.	Two or more machines with the same IP address exist in the CloudLink Center.	High	Ensure that you set a unique IP address for one of the machines.
37	New key is generated to comply to higher security requirements. Change the passcodes to use the new key	New key is generated to comply with higher security requirements. This feature is supported in CloudLink 7.1 and later versions.	CloudLink is upgraded from earlier versions to 7.1 version.	High	Ensure that you update the Vault passcodes to use the new key.
38	CloudLink Center is unable to synchronize its time with the NTP server	CloudLink Center is unable to synchronize its time with the NTP server. You can synchronize CloudLink Center with the date and time that is obtained from Network Time Protocol (NTP) servers. By default, CloudLink Center is configured with four global NTP servers.	This alarm is triggered when the: • Number of configured NTP servers is zero. • NTP server is unable to synchronize its time with CloudLink Center.	Low	To clear this alarm: • Configure at least one NTP server. • Force an NTP server time synchronization with CloudLink Center.
39	SNMP target host is deleted	An SNMP notification target host that was configured is deleted. CloudLink Center sends SNMP traps to the	The configured SNMP notification target host is deleted.	Low	This alarm is cleared after few minutes. No action required.

Table 2. CloudLink Center alarms and relevant actions (continued)

Sl. No.	Alarm	Description	Cause	Severity	Action
		configured target host when CloudLink alarms are triggered, updated, or lowered.			
40	The cluster node certificate is expired.	The cluster node certificate is expired.	The existing cluster node certificate has expired.	High	Ensure that you upload a new and valid cluster node certificate.
41	The web TLS certificate is expired.	The web TSL certificate is expired.	The existing web TLS certificate has expired.	High	Ensure that you upload a new and valid web TLS certificate.
42	Unsupported encryption policy for SDS.	The **Boot and Manual Data** and **Boot and All Data** encryption policies are not supported on the SDS drives installed on a PowerFlex device. Only the **Manual** and **All Data** encryption policies are supported on the SDS drives installed on PowerFlex.	Only for storage node drives on a PowerFlex, the aforesaid operations cannot be performed because the policies are not supported.	Low	No recommended action. Ensure that you do not perform the aforesaid operations on SDS drives installed on PowerFlex.
43	Device mapper update	The SED or SDS device mapper is not updated.	The SED or SDS device mapper is not updated.	High	Ensure that you update the device mapper for all drives.
44	Restart CloudLink services	Restart CloudLink services after uploading the cluster CA certificate.	Cluster synchronization uses a third-party symmetricDS library, which runs as an embedded service on Cloudlink application. After the cluster certificate change, the symmetricDS library fails to reload the new certificate.	High	Contact the Dell Technologies Support team to restart CloudLink services.
45	Machines are disconnected	One or more machines are disconnected.	One or more machines are disconnected.	Low	This alarm is cleared after the CloudLink agents are connected. No recommended action.